

Problem-Solution Fit

Date	29 June 2026
Team ID	N/A
Project Name	OptiCrop Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

Problem–Solution Fit Canvas – OptiCrop

• Small & medium-scale farmers • Agricultural officers • Agribusiness companies • Agricultural researchers & policymakers	• Wrong crop selection – High • Low crop productivity – High • Unpredictable weather – High • Lack of scientific guidance – Medium
• Crop failure • Low profits • Rising fertilizer costs • Seasonal weather changes	Before: Confused, stressed, uncertain. After: Confident, informed, optimistic, satisfied.
• Traditional farming advice • Local agriculture experts • Weather apps Pros: Experienced guidance. Cons: Generic, slow, not personalized.	• Limited budget • Basic smartphones • Poor internet in rural areas • Limited technical knowledge
• Checks weather regularly • Consults neighboring farmers • Uses agriculture apps occasionally • Depends on seasonal experience	ONLINE: • Website • Mobile App • WhatsApp • Social Media OFFLINE: • Agriculture offices • Farmer meetings • Local markets • Training programs

• Poor soil analysis • Climate variability • Lack of data-driven decisions • Limited access to modern agricultural technology	OptiCrop is an AI-powered agricultural production optimization engine that analyzes soil nutrients (N, P, K), temperature, humidity, pH, and rainfall to recommend the most suitable crop, assess crop suitability, and support data-driven farming, research, and policy decisions for higher productivity and sustainable agriculture.
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